PES-VCS Lab Report

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Repository: https://github.com/samanvithkashyap/PES2UG24CS432-PES-VCS

Phase 1

Screenshot 1A:

```
samanvith@Sam:~/Desktop/clg/OS/orange2/os-u4-orange-problem$ make test_objects
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -Wall -Wextra -O2 -c object.c -o object.o
gcc -o test_objects test_objects.o object.o -lcrypto
samanvith@Sam:~/Desktop/clg/OS/orange2/os-u4-orange-problem$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b
9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe
9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
samanvith@Sam:~/Desktop/clg/OS/orange2/os-u4-orange-problem$
```

Screenshot 1B:

```
samanvith@Sam:~/Desktop/clg/OS/orange2/os-u4-orange-problem$ find .pes/objects -
type f
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
samanvith@Sam:~/Desktop/clg/OS/orange2/os-u4-orange-problem$
```

Phase 2

Screenshot 2A:

```
samanvith@Sam:~/Desktop/clg/OS/orange2/os-u4-orange-problem$ make test_tree
gcc -Wall -Wextra -O2 -c test_tree.c -o test_tree.o
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
gcc -o test_tree test_tree.o object.o tree.o index.o -lcrypto
samanvith@Sam:~/Desktop/clg/OS/orange2/os-u4-orange-problem$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
samanvith@Sam:~/Desktop/clg/OS/orange2/os-u4-orange-problem$
```

Screenshot 2B:

```
samanvith@Sam:~/Desktop/clg/OS/orange2/os-u4-orange-problem$ xxd .pes/objects/a6/882ee4afdc855ea4e39f4e0fc77d90e73a6f72deb383e11934310a10a030f6 | head -20
00000000: 7472 6565 2039 3600 3130 3036 3434 2062  tree 96.100644 b
00000010: 7965 2e74 7874 00e6 7ed6 6bcb 4a70 8250  ye.txt..~.k.Jp.P
00000020: a89f 315d 4d8b b927 0334 3c84 df83 4438  ..1]M..'.4<...D8
00000030: 880e 13a6 8652 bc31 3030 3634 3420 6865  .....R.100644 he
00000040: 6c6c 6f2e 7478 7400 05c8 d777 d0be 8574  llo.txt....w...t
00000050: 8b67 88d1 b5ca f7d6 9754 e6e6 8060 c265  .g.......T...`.e
00000060: ca21 cbd6 c05e 5f43                      .!...^_C
samanvith@Sam:~/Desktop/clg/OS/orange2/os-u4-orange-problem$
```

Phase 3

Screenshot 3A:

```
samanvith@Sam:~/Desktop/clg/OS/orange2/os-u4-orange-problem$ ./pes init
Initialized empty PES repository in .pes/
samanvith@Sam:~/Desktop/clg/OS/orange2/os-u4-orange-problem$ echo "hello" > file1.txt
samanvith@Sam:~/Desktop/clg/OS/orange2/os-u4-orange-problem$ echo "world" > file2.txt
samanvith@Sam:~/Desktop/clg/OS/orange2/os-u4-orange-problem$ ./pes add file1.txt file2.txt
samanvith@Sam:~/Desktop/clg/OS/orange2/os-u4-orange-problem$ ./pes status
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  test_sequence.sh
  untracked:  index.h
  untracked:  tree.h
  untracked:  .gitignore
  untracked:  test_objects.c
  untracked:  pes.h
  untracked:  index.c
  untracked:  pes.c
  untracked:  object.c
  untracked:  commit.h
  untracked:  test.txt
  untracked:  test_tree.c
  untracked:  commit.c
  untracked:  README.md
  untracked:  Makefile
  untracked:  tree.c
```

Screenshot 3B:

```
samanvith@Sam:~/Desktop/clg/OS/orange2/os-u4-orange-problem$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776752437 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776752440 6 file2.txt
samanvith@Sam:~/Desktop/clg/OS/orange2/os-u4-orange-problem$
```

Phase 4

Screenshot 4A:

```
samanvith@Sam:~/Desktop/clg/OS/orange2/os-u4-orange-problem$ rm -rf .pes
samanvith@Sam:~/Desktop/clg/OS/orange2/os-u4-orange-problem$ ./pes init
Initialized empty PES repository in .pes/
samanvith@Sam:~/Desktop/clg/OS/orange2/os-u4-orange-problem$ echo "Hello" > hello.txt
samanvith@Sam:~/Desktop/clg/OS/orange2/os-u4-orange-problem$ ./pes add hello.txt
samanvith@Sam:~/Desktop/clg/OS/orange2/os-u4-orange-problem$ ./pes commit -m "Initial commit"
Committed: de0521b6f57a... Initial commit
samanvith@Sam:~/Desktop/clg/OS/orange2/os-u4-orange-problem$ echo "World" >> hello.txt
samanvith@Sam:~/Desktop/clg/OS/orange2/os-u4-orange-problem$ ./pes add hello.txt
samanvith@Sam:~/Desktop/clg/OS/orange2/os-u4-orange-problem$ ./pes commit -m "Add world"
Committed: f20759745c3b... Add world
samanvith@Sam:~/Desktop/clg/OS/orange2/os-u4-orange-problem$ echo "Goodbye" > bye.txt
samanvith@Sam:~/Desktop/clg/OS/orange2/os-u4-orange-problem$ ./pes add bye.txt
samanvith@Sam:~/Desktop/clg/OS/orange2/os-u4-orange-problem$ ./pes commit -m "Add farewell"
Committed: 4a7bd7e30123... Add farewell
samanvith@Sam:~/Desktop/clg/OS/orange2/os-u4-orange-problem$ ./pes log
commit 4a7bd7e30123006893bbb2ada646b5a4e59e2fe738607698b4dcccb280e8fef9
Author: Samanvith Kashyap <PES2UG24CS432>
Date:   1776753289

    Add farewell

commit f20759745c3b1ee40c91dace01b07ecf211b3713f30d944624feaf303d003241
Author: Samanvith Kashyap <PES2UG24CS432>
Date:   1776753279

    Add world

commit de0521b6f57a97eeefe75be8a0960e3b291a4398e08d001f001d01e91d923060
Author: Samanvith Kashyap <PES2UG24CS432>
Date:   1776753269

    Initial commit

samanvith@Sam:~/Desktop/clg/OS/orange2/os-u4-orange-problem$
```

Screenshot 4B:

```
samanvith@Sam:~/Desktop/clg/OS/orange2/os-u4-orange-problem$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/4a/7bd7e30123006893bbb2ada646b5a4e59e2fe738607698b4dcccb280e8fef9
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/a6/882ee4afdc855ea4e39f4e0fc77d90e73a6f72deb383e11934310a10a030f6
.pes/objects/bd/9b769bb4d9910c31b4f0e99e4375fffea35c2055130c248ff131f78c626442
.pes/objects/d8/f28203071e10a3bde605928368bc5b51871287a867d8d10382967099dde814
.pes/objects/de/0521b6f57a97eeefe75be8a0960e3b291a4398e08d001f001d01e91d923060
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/objects/f2/0759745c3b1ee40c91dace01b07ecf211b3713f30d944624feaf303d003241
.pes/refs/heads/main
samanvith@Sam:~/Desktop/clg/OS/orange2/os-u4-orange-problem$
```

Screenshot 4C:

```
samanvith@Sam:~/Desktop/clg/OS/orange2/os-u4-orange-problem$ echo "=== .pes/HEAD ==="
=== .pes/HEAD ===
samanvith@Sam:~/Desktop/clg/OS/orange2/os-u4-orange-problem$ cat .pes/HEAD
ref: refs/heads/main
samanvith@Sam:~/Desktop/clg/OS/orange2/os-u4-orange-problem$ echo "=== .pes/refs/heads/main
=== .pes/refs/heads/main ===
samanvith@Sam:~/Desktop/clg/OS/orange2/os-u4-orange-problem$ cat .pes/refs/heads/main
4a7bd7e30123006893bbb2ada646b5a4e59e2fe738607698b4dcccb280e8fef9
samanvith@Sam:~/Desktop/clg/OS/orange2/os-u4-orange-problem$
```

Integration test:

```
--- Third Commit ---
Committed: 4b08207e11e8... Add farewell

--- Full History ---
commit 4b08207e11e89ff7a4e9d835983b2c9aab336e20ec7d7088da7b439e06f85719
Author: Samanvith Kashyap <PES2UG24CS432>
Date:   1776753453

    Add farewell

commit 0219d30f920a3a98851627318583050a344c741b960645108117728c4d4a9ef2
Author: Samanvith Kashyap <PES2UG24CS432>
Date:   1776753453

    Update file.txt

commit 5d3a00cc424fa383a829f466dae43cb17b97b2f629783dcfded3e79281e65590
Author: Samanvith Kashyap <PES2UG24CS432>
Date:   1776753453

    Initial commit


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
4b08207e11e89ff7a4e9d835983b2c9aab336e20ec7d7088da7b439e06f85719

--- Object Store ---
Objects created:
10
.pes/objects/02/19d30f920a3a98851627318583050a344c741b960645108117728c4d4a9ef2
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/4b/08207e11e89ff7a4e9d835983b2c9aab336e20ec7d7088da7b439e06f85719
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/5d/3a00cc424fa383a829f466dae43cb17b97b2f629783dcfded3e79281e65590
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff

=== All integration tests completed ===
samanvith@Sam:~/Desktop/clg/OS/orange2/os-u4-orange-problem$
```

Phase 5

Q5.1

To implement `pes checkout <branch>`, `.pes/HEAD` is rewritten to contain `ref: refs/heads/<branch>`, and `.pes/index` is rebuilt to match the target branch's tree. The working directory must be updated: files in the target tree are written out, files only in the current tree are deleted, and unchanged files are left alone. It's complex because it's not atomic — writing many files can fail midway, leaving the working directory inconsistent. It must also detect uncommitted changes before starting to avoid destroying user work, and handle edge cases like case-insensitive filesystems, symlinks, and paths that change from file to directory between branches.

Q5.2

For each entry in the index, `stat()` the path. If `mtime` and `size` match the index values, the file is unchanged. Otherwise, read the file, compute its blob hash, and compare against the index entry's hash. If the hash differs, the file is dirty. Checkout must refuse when a dirty file has a different hash in the target branch's tree than in the index, because switching would overwrite the uncommitted change.

Q5.3

In detached HEAD, `.pes/HEAD` contains a commit hash directly instead of `ref: refs/heads/<name>`. Commits made in this state update HEAD to point at the new commit, but no branch references it. Switching to any branch moves HEAD and leaves the new commits orphaned — reachable by nothing, eligible for garbage collection. To recover, note the current HEAD hash before switching (`cat .pes/HEAD`), then after switching, create a new branch pointing at that hash by writing it into `.pes/refs/heads/<recovery-branch>`.

Phase 6

Q6.1

Use mark-and-sweep. Start with all branch tips in `.pes/refs/heads/` plus HEAD. For each commit, mark its tree and parent reachable; for each tree, mark its blobs and subtrees reachable. Repeat until nothing new is added. Use a hash set of object IDs for O(1) lookup. Then walk `.pes/objects/` and delete any file whose hash isn't in the set. For 100,000 commits $\times$ 50 branches, deduplication means shared objects aren't re-counted — you'd visit roughly 700,000 objects total (100,000 commits plus a few new trees and blobs per commit), not 100,000 $\times$ 50.

Q6.2

A `commit` operation writes tree and blob objects before writing the commit object that references them. If GC runs in that window, those new trees and blobs aren't yet reachable from any ref, so GC marks them unreachable and deletes them. The commit then finishes and writes a commit object pointing to deleted trees — the repo is corrupted. Git avoids this by only pruning objects older than a grace period (default two weeks), so freshly written loose objects are always safe. It also uses locks and reflog entries as additional reachability roots.